

Adaptive Retrieval-Augmented Generation with Dynamic Confidence Thresholds for Domain-Specific Question Answering

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Abstract

We present AdaptRAG, a novel framework for retrieval-augmented generation that dynamically adjusts confidence thresholds based on query complexity and domain specificity. Unlike static RAG systems that apply uniform retrieval strategies, AdaptRAG employs a learned routing mechanism that classifies queries into complexity tiers (simple factual, multi-hop reasoning, and open-ended analytical) and adjusts the number of retrieved passages, re-ranking strategy, and generation parameters accordingly. We evaluate AdaptRAG on five domain-specific benchmarks spanning legal, medical, financial, scientific, and technical domains. Our approach achieves state-of-the-art results on all benchmarks, improving answer accuracy by 8.3% on average while reducing retrieval latency by 34% compared to existing methods. We further demonstrate that adaptive confidence thresholds reduce hallucination rates by 41% on adversarial evaluation sets. Code and models are available at github.com/stanford-nlp/adaptrag.

1. Introduction

Retrieval-Augmented Generation (RAG) has emerged as the predominant paradigm for building knowledge-intensive NLP systems that combine the parametric knowledge of large language models (LLMs) with non-parametric retrieval from external knowledge bases (Lewis et al., 2020; Borgeaud et al., 2022). However, current RAG implementations suffer from a fundamental limitation: they apply uniform retrieval and generation strategies regardless of query complexity.

Consider two queries in a medical domain: "What is the half-life of amoxicillin?" requires simple factual retrieval from a single passage, while "Compare treatment protocols for stage III non-small cell lung cancer in elderly patients with comorbidities" demands multi-hop reasoning across multiple documents with careful evidence synthesis. Applying the same retrieval pipeline to both queries leads to either over-retrieval (wasting compute and introducing noise for simple queries) or under-retrieval (missing critical context for complex queries).

In this paper, we introduce AdaptRAG, which addresses this limitation through three key contributions:

- 1) A query complexity classifier trained on 50,000 annotated queries across five professional domains
- 2) A dynamic threshold mechanism that adjusts retrieval depth (k), re-ranking aggressiveness, and chunk granularity based on predicted complexity
- 3) A confidence-aware generation module that calibrates output certainty and triggers additional retrieval when confidence falls below domain-specific thresholds

2. Related Work

RAG Systems: The original RAG framework (Lewis et al., 2020) demonstrated that combining retrieval with generation improves factual accuracy. Subsequent work has explored dense passage retrieval (Karpukhin et al., 2020), iterative retrieval (Trivedi et al., 2023), and self-reflective retrieval (Asai et al., 2024). Our work differs by introducing adaptive complexity-based routing.

Confidence Estimation in NLP: Calibration of neural models has been studied extensively (Guo et al., 2017; Desai & Durrett, 2020). Recent work on LLM uncertainty estimation (Kadavath et al., 2022; Kuhn et al., 2023) provides foundations for our confidence-aware generation module.

Query Complexity: Prior work has classified queries by difficulty (Fan et al., 2019; Mallen et al., 2023), but primarily for routing between retrieval-augmented and parametric-only generation. We extend this to fine-grained control over the entire RAG pipeline.

3. Methodology

3.1 Query Complexity Classification

We define three complexity tiers based on the reasoning structure required:

Tier 1 - Simple Factual (SF): Single-hop retrieval, direct answer extraction. Example: "What year was GDPR enacted?"

Tier 2 - Multi-hop Reasoning (MHR): Requires synthesizing information from 2-5 passages with logical chaining. Example: "How does metformin interact with contrast dye in patients with reduced GFR?"

Tier 3 - Analytical Synthesis (AS): Open-ended analysis requiring comprehensive evidence gathering, comparison, and nuanced reasoning. Example: "Evaluate the pros and cons of transformer-based vs. graph neural network approaches for molecular property prediction."

Our classifier is a fine-tuned DeBERTa-v3-large model achieving 94.2% accuracy on a held-out test set of 5,000 queries.

3.2 Dynamic Threshold Mechanism

For each complexity tier t in {SF, MHR, AS}, we define:

- Retrieval depth $k(t)$: SF=3, MHR=8, AS=15
- Re-ranking threshold $\tau(t)$: SF=0.7, MHR=0.5, AS=0.3
- Chunk size $c(t)$: SF=256 tokens, MHR=512 tokens, AS=1024 tokens
- Confidence threshold $\theta(t)$: SF=0.9, MHR=0.75, AS=0.6

These are initial values learned through Bayesian optimization on validation sets and further adapted per-domain.

4. Results

Table 1: Performance comparison on domain-specific benchmarks (accuracy %)

	Legal	Medical	Financial	Scientific	Technical	Avg
Vanilla RAG	71.2	68.4	73.1	69.8	72.5	71.0
Self-RAG	74.8	72.1	76.3	73.2	75.9	74.5
CRAG	76.1	74.5	77.8	75.0	77.2	76.1
AdaptRAG (ours)	82.4	81.2	84.1	80.7	83.5	82.4

AdaptRAG achieves consistent improvements across all domains, with the largest gains in medical (+6.7%) and financial (+6.3%) domains where query complexity varies most significantly.

Hallucination Rate (adversarial set):

Vanilla RAG: 23.4%, Self-RAG: 18.2%, CRAG: 15.7%, AdaptRAG: 9.3%

Latency (avg. ms per query):

Vanilla RAG: 2,340ms, Self-RAG: 3,120ms, CRAG: 2,890ms, AdaptRAG: 1,540ms

5. Conclusion

We presented AdaptRAG, a framework that dynamically adjusts RAG pipeline parameters based on query complexity. By routing simple queries through lightweight retrieval and reserving deep retrieval for complex analytical queries, AdaptRAG achieves both improved accuracy and reduced latency. The confidence-aware generation module further

reduces hallucinations by 41% through calibrated uncertainty estimation and iterative retrieval triggers.

Future work will explore: (1) extending AdaptRAG to multimodal retrieval, (2) online learning of threshold parameters, and (3) application to conversational multi-turn QA settings.